Chi-Square Tests & Regression Inference

Section 1: Chi-Square Goodness-of-Fit Test

1 What is the formula for the Chi-Square test statistic?

a. $\chi^2 = \sum \frac{(O-E)^2}{E}$

b. $\chi^2 = \sum \frac{(O-E)}{E}$

c. $\chi^2 = \sum \frac{(O-E)^2}{O}$

d. $\chi^2 = \frac{(\bar{x}-\mu)^2}{s^2}$

2 A bag of candies claims to have 5 colors in equal proportions (20% each). In a sample of 100 candies, what are the degrees of freedom for the goodness-of-fit test?

a. 4

b. 5

c. 99

d. 100

3 What condition must be satisfied regarding expected counts for a Chi-Square test to be valid?

a. All expected cell counts must be at least 5

b. All observed cell counts must be at least 30

c. The sample size must be exactly 100

d. The categories must be normally distributed

Section 2: Chi-Square Test of Independence vs. Homogeneity

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4 In a two-way contingency table with r rows and c columns, how is the expected count for each cell calculated under H_0?

a. $E = \frac{\text{Row Total} \times \text{Column Total}}{\text{Table Total}}$

b. $E = \frac{\text{Row Total} + \text{Column Total}}{2}$

c. $E = \frac{\text{Table Total}}{r \times c}$

d. $E = \frac{\text{Row Total}}{\text{Column Total}}$

5 A single random sample of 300 voters is categorized by political affiliation and stance on a ballot issue. Which test assesses if affiliation and stance are associated?

a. Chi-Square Test of Independence

b. Chi-Square Test of Homogeneity

c. Chi-Square Goodness-of-Fit Test

d. Two-sample z-test for proportions

6 What are the degrees of freedom for a Chi-Square test on a 3×4 contingency table?

a. 6

b. 12

c. 7

d. 11

Section 3: Inference for Linear Regression Slope

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7 In testing for a significant linear relationship between x and y, what are the standard null and alternative hypotheses for the population slope β?

a. $H_0 : \beta = 0$ versus $H_a : \beta \neq 0$

b. $H_0 : \beta = 1$ versus $H_a : \beta \neq 1$

c. $H_0 : b = 0$ versus $H_a : b \neq 0$

d. $H_0 : r = 1$ versus $H_a : r \neq 1$

8 Computer regression output shows: Predictor Coef SE Coef T P; Constant 14.2 2.1 6.76 0.000; Slope 3.45 0.69 5.00 0.0001. For $n = 18$ observations, what is the t-statistic for the slope?

a. 5.00

b. 6.76

c. 3.45

d. 0.69

9 What are the degrees of freedom for the t-test of the slope in simple linear regression with n data pairs?

a. $n - 2$

b. $n - 1$

c. n

d. $2n - 2$

Section 4: Regression Model Conditions (LINER)

10 What does the acronym LINER stand for regarding the conditions for regression inference?

a. Linear, Independent, Normal y for each x, Equal variance, Random sampling

b. Least squares, Intercept, Normal, Errors, Residuals

c. Linear, Integrated, Numerical, Estimated, Robust

d. Limit, Interval, Normal, Expected, Ratio

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11 If a residual plot displays a funnel or cone shape (spread widens as x increases), which condition is violated?

a. Equal variance of residuals (homoscedasticity)

b. Linearity

c. Independence

d. Random sampling

12 A 95% confidence interval for the regression slope β is $(1.24, 3.86)$. What can be concluded at $\alpha = 0.05$?

a. Because 0 is not in the interval, there is convincing evidence of a positive linear relationship between x and y

b. Because 0 is not in the interval, the relationship is non-linear

c. The correlation coefficient r must equal 1.24

d. We fail to reject $H_0 : \beta = 0$

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